

# Matthew Maloni

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## EDUCATION

### University of Florida, Herbert Wertheim College of Engineering

*M.S. in Electrical and Computer Engineering*

*Incoming Spring 2027*

Gainesville, FL

### University of Florida, College of Liberal Arts and Sciences

*Bachelor of Science in Data Science*

*August 2023 – December 2026*

Gainesville, FL

GPA: 3.62 / 4.00 · Dean's List: Fall 2023, Fall 2025

**Relevant Coursework:** Applied Machine Learning Systems; Data Structures & Algorithms; Linear Algebra for Data Science; Probability; Linear Regression; Programming with Data in R; Digital Logic; Information and Database Systems

## WORK EXPERIENCE

### Kore Capital

*May 2026 – July 2026*

*Financial Engineering Intern*

Boynton Beach, FL

- Co-developed a linear-regression equity factor model spanning momentum, value, quality, size, and growth baskets, applying  $R^2$  and autocorrelation testing on Bloomberg data to evaluate hedge quality and factor attribution.
- Integrated Claude MCP servers into the firm's .NET portfolio system and Excel add-in, condensing tax-lot, PnL, and reporting workflows into automated tools that surface substrategy-level analytics.
- Consolidated 100K+ position, tax-lot, and trade records into a unified analytics layer, building per-campaign visualizations that expose trading patterns and adverse-move risk.

### Pastry Art Wine Club

*October 2025 – Present*

*Software Developer*

Sarasota, FL (Remote)

- Designed and built a production iOS app for a wine retailer's subscription club, from initial requirements through App Store review and release, fixing production bugs and shipping updates against real customer usage.
- Built a wine-label scanner that reads a user's photo and matches it to the store's catalog in a single AI call, and a recommendation chatbot that builds user memories from how they interact with the app and its AI.

### Trustwell

*April 2025 – August 2025*

*Data Science Intern*

Salem, OR (Remote)

- Compiled multi-region regulatory datasets and integrated the OpenAI API to improve response quality for Trustwell's food-safety compliance assistant; supported the frontend team's AngularJS-to-Vue-3 migration.

## RESEARCH & LEADERSHIP

### IEEE Signal Processing Society, UF Student Chapter

*September 2025 – Present*

*Research Board, Project Lead*

Gainesville, FL

- Implemented scoring and trust-gating algorithms for Plato's Cave, a knowledge-graph paper-verification system aggregating paper-level integrity from six graph metrics, including bridge coverage, best-path reliability, max-flow redundancy, and min-cut fragility; curated a 104-PDF evaluation dataset and co-authored the resulting paper, accepted to the AAAI Spring Symposium.
- Founded and lead a five-member research group building binaural (dual-microphone) audio scene analysis for real-world sound-source localization and separation.

### Nettles Lab, UF Scripps Institute

*January 2026 – May 2026*

*Undergraduate Researcher*

Jupiter, FL

- Trained a VAMPnet in PyTorch mapping molecular dynamics trajectories to metastable conformational states for NIH-funded glucocorticoid receptor drug discovery, reaching a VAMP-2 score of 2.87 against a max of 3.0.

### Beta Theta Pi, Gamma Xi Chapter

*August 2025 – Present*

*Internal Philanthropy Chair*

Gainesville, FL

- Directed Beta Bring It On 2026, raising a record \$64,000 for the Alzheimer's Association and selling out the Phillips Center's 1,700 seats with 300+ participants across 19 teams.

## PROJECTS

### PortalPath: Graph-Based Ranking for the CFB Transfer Portal

March 2026 · Team of 3 · Python, C++, Dash

- Built an end-to-end pipeline over the CollegeFootballData API condensing 726,000 raw records into 75,492 clean player-seasons; designed a PageRank-based prestige-diffusion model over a 5,201-edge transfer graph ranking 132 programs.

### CNN Medical Image Classification

November 2025 · TensorFlow/Keras, HiPerGator

- Compared four CNN architectures on chest X-ray and diabetic retinopathy classification, achieving 94.77% accuracy and 83.61% AUC across 14 thoracic pathologies over 78,468 images.

## SKILLS

**Languages:** Python, Java, C++, R, SQL, MATLAB, Visual Basic, HTML/CSS, Bash

**Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, deeptime, CNNs, VAMPnets/Markov state models, self-supervised learning, random forests, k-means, regularized regression, cross-validation, PCA/ISOMAP, feature engineering

**AI Systems:** OpenAI API, Claude/MCP, LLM agent design, retrieval-augmented pipelines, vector databases, knowledge graphs

**Infrastructure & Computing:** AWS, PostgreSQL, Supabase, Docker, Git/GitHub, HPC (SLURM, HiPerGator)